APB Master Verification Report

Overall Code Coverage: 85.55%

Overview

A SystemVerilog class-based verification environment was developed to verify the APB Master. The environment consists of a generator, driver, monitor, reference model, scoreboard, environment, interface and top-level testbench. Directed and constrained-random tests were executed to verify APB protocol functionality.

DUT Bug Identified

According to the design specification, after the ACCESS state completes:
• If transfer = 0 and PREADY = 1, the FSM should transition to IDLE.
• If transfer = 1, the FSM should transition directly to SETUP to support back-to-back transfers.
During verification it was observed that the DUT did not transition to SETUP for consecutive transfers, resulting in incorrect handling of back-to-back APB transactions.

Coverage Notes for Uncovered Items

- Overall code coverage achieved is 85.55%.

Questa Branch Coverage Report

Show All | Show Covered | Show Missing

if(exp_trans.write_read==1 && act_trans.write_read==1) begin — 66.66%

Branch	Source	Hits	Status
IF	if(exp_trans.write_read==1 && act_trans.write_read==1) begin	23	Covered
TRUE	else if(act_trans.write_read==0 && exp_trans.write_read==0) begin	7	Covered
ELSE	else begin	0	ZERO

if((exp_trans.PSEL==act_trans.PSEL) && (exp_trans.PENABLE== act_trans.PEN... — 50.00%

Branch	Source	Hits	Status
IF	if((exp_trans.PSEL==act_trans.PSEL) && (exp_trans.PENABLE== act_trans.PEN...	23	Covered
ELSE	else begin	0	ZERO

if((act_trans.PSEL==exp_trans.PSEL) && (act_trans.PENABLE==exp_trans.PENA... — 50.00%

Branch	Source	Hits	Status
IF	if((act_trans.PSEL==exp_trans.PSEL) && (act_trans.PENABLE==exp_trans.PENA...	7	Covered
ELSE	else begin	0	ZERO

- Scoreboard coverage is lower because the uncovered branches correspond to error-handling (else) paths. During normal verification, the DUT output matched the reference model, so mismatch branches were intentionally not executed.

- Some scoreboard branches are only exercised when expected and actual transactions differ. Since no mismatches occurred during normal tests, these branches remain uncovered.

Questa Condition Coverage Report

Show All | Show Covered | Show Missing

FEC Condition: (vif.mon_cb.PSEL && vif.mon_cb.PENABLE && ~vif.mon_cb.PREADY)			33.33%
Input Term	**Covered**	**Reason For No Coverage**	**Hint**
vif.mon_cb.PSEL	No	'_0' not hit	Hit '_0'
vif.mon_cb.PENABLE	No	'_0' not hit	Hit '_0'
vif.mon_cb.PREADY	Yes		
Rows	**FEC Target**	**Hits**	**Non-Masking Condition(s)**
Row 1	vif.mon_cb.PSEL_0	0	-
Row 2	vif.mon_cb.PSEL_1	1	(vif.mon_cb.PENABLE && ~vif.mon_cb.PREADY)
Row 3	vif.mon_cb.PENABLE_0	0	vif.mon_cb.PSEL
Row 4	vif.mon_cb.PENABLE_1	1	(vif.mon_cb.PSEL && ~vif.mon_cb.PREADY)
Row 5	vif.mon_cb.PREADY_0	1	(vif.mon_cb.PSEL && vif.mon_cb.PENABLE)
Row 6	vif.mon_cb.PREADY_1	1	(vif.mon_cb.PSEL && vif.mon_cb.PENABLE)

- In the monitor, one FEC condition is not obtainable because sampling occurs only after waiting for both PSEL and PENABLE. At that point the protocol is already in the ACCESS state, and the monitor counts wait cycles until PREADY becomes high.

Scope: /apb_pkg/apb_driver
Covergroup type: drv_cg

Cross: wr_strb

Crossed Coverpoints : wr_rd , strb

Search:

wr_rd	strb	At Least	Hits
read	all_ones_strb	1	0
read	single_bit_strb	1	0
write	zero_strb	1	3
read	zero_strb	1	7
write	all_ones_strb	1	2
write	single_bit_strb	1	4

- A cross-coverage bin involving read/write and PSTRB cannot be hit because APB read transactions do not use write strobes. The verification environment also constrains read transactions so that PSTRB is not asserted.

Scope: /apb_pkg/apb_monitor
Covergroup type: mon_cg

Cross: wr_str

Crossed Coverpoints : wr_rd , str

Search:

Bin Name	At Least	Hits
write_strobe	1	7

Search:

wr_rd	str	At Least	Hits
read	ones_strb1	1	0
read	zero_strb1	1	7

- Similarly, monitor bins involving read transactions with valid strobe values cannot be covered because such protocol combinations are illegal.

case (state) — 75.00%

Branch	Source	Hits	Status
TRUE	IDLE: begin	50	Covered
TRUE	SETUP: begin	47	Covered
TRUE	ACCESS: begin	48	Covered
TRUE	default: next_state = IDLE;	0	ZERO

if (transfer) — 100.00%

Branch	Source	Hits	Status
IF	if (transfer)	48	Covered

- In the DUT, one branch cannot be covered because the FSM state is encoded as a 2-bit signal and all legal state values are already explicitly handled in the case statement. There is no remaining legal branch to exercise.

Test Cases Executed

- Reset
- Single Write
- Single Read
- Back-to-Back Transfers
- Random Read/Write
- Random Address

Coverage Screenshots

Questa Coverage Report

Number of tests run:	1
Passed:	0
Warning:	1
Error:	0
Fatal:	0

List of tests included in report...

List of global attributes included in report...

List of Design Units included in report...

Coverage Summary by Structure:

Design Scope	Hits %	Coverage %
top	98.98%	96.19%
intrf	99.14%	98.12%
DUV	99.03%	96.34%
apb_pkg	82.74%	65.84%
apb_transaction/copy	100.00%	100.00%
apb_generator/new	100.00%	100.00%
apb_generator/start	100.00%	100.00%
apb_driver	74.46%	81.26%
apb_monitor	94.23%	76.19%
apb_scoreboard/new	100.00%	100.00%
apb_scoreboard/start	100.00%	100.00%
apb_scoreboard/compare_report	28.94%	33.76%
apb_environment/new	100.00%	100.00%
apb_environment/build	100.00%	100.00%
apb_environment/start	100.00%	100.00%
apb_test/new	100.00%	100.00%
apb_test/run	100.00%	100.00%

Coverage Summary by Type:

Coverage Type	Bins	Hits	Misses	Weight	% Hit	Coverage
Total Coverage:					94.76%	85.55%
Covergroups	42	39	3	1	92.85%	95.95%
Statements	308	276	32	1	89.61%	89.61%
Branches	44	39	5	1	88.63%	88.63%
FEC Conditions	28	12	16	1	42.85%	42.85%
Toggles	754	749	5	1	99.33%	99.33%
FSMs	7	6	1	1	85.71%	87.50%
States	3	3	0	1	100.00%	100.00%
Transitions	4	3	1	1	75.00%	75.00%
Assertions	20	19	1	1	95.00%	95.00%

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vcover report -html apb_coverage.ucdb -htmldir covReport -details